

Activity-1

1. Medical Image Enhancement

Q1. Which gray-level transformation would be recommended?

Log Transformation

Q2. Why is it suitable?

It enhances the darker region of the x-ray, making soft tissue more visible while keeping bright bone regions clear.

Q3. What effect will it have on the image?
The dark tissues become brighter and more detailed, improving the overall visibility of the x-ray.

2. Night Time Surveillance

Q1. Which transformation would improve the visibility?

Log Transformation

Q2. Why is it more suitable than Negative transformation?

Log transformation increases the brightness of dark pixels whereas negative transformation only inverts the image and does not improve visibility.

Q3 predict the appearance of the enhanced image

The image will become brighter, with objects like vehicles, people, and the surroundings appearing more visible and easier to recognize.

3. Fingerprint Recognition

Q1 Which gray-level transformation would you apply?

Contrast Stretching (piecewise Linear Transformation)

Q2 Explain why

It increases the contrast between fingerprint ridges and valleys, making the fine ridge details clearer for accurate recognition.

Q3 What would happen if a log transformation were applied instead?

Log transformation would brighten the image but would not effectively improve the ridge-to-valley contrast. So fingerprint details may remain unclear.